Solution for Chapter 5

Zimo Guo

5.1

(a) When $p = 2$,

$$\mathcal{L}(\mathbf{W}, b, \alpha, \beta, \xi) = \frac{1}{2}||\mathbf{w}||^2 + C\sum_{i=1}^{m}\xi_i^2 - \sum_{i=1}^{m}\alpha_i[y_i(\mathbf{w}\mathbf{x}_i + b) - 1 + \xi_i] - \sum_{i=1}^{m}\beta_i\xi_i$$

Setting the gradient of the Lagrangian with respect to the $\mathbf{w}, b, \xi_i$ is to zero, the results we got is same as (5.26), (5.27), (5.29), (5.30). (5.28) is replaced by $2C\xi_i = \alpha_i + \beta_i$.

$$\begin{aligned}\inf \mathcal{L} &= \frac{1}{2}||\sum_{i=1}^{m}\alpha_i y_i \mathbf{x}_i||^2 + C\sum_{i=1}^{m}(\frac{\alpha_i + \beta_i}{2C})^2 - \sum_{i=1}^{m}\alpha_i y_i b + \sum_{i=1}^{m}\alpha_i - \sum_{i=1}^{m}\alpha_i\xi_i - \sum_{i=1}^{m}\beta_i\xi_i - \sum_{i=1}^{m}\alpha_i y_i \mathbf{w}\mathbf{x} \\ &= \sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\alpha_i\alpha_j y_i y_j \mathbf{x}_i\mathbf{x}_j - \frac{1}{4}\sum_{i=1}^{m}\frac{(\alpha_i + \beta_i)^2}{C}\end{aligned}$$

(b) Yes, this problem is still convex because the first two items are convex, and $-\frac{1}{4}\sum_{i=1}^{m}\frac{(\alpha_i+\beta_i)^2}{C}$ is concave but we can use convec techniques to solve it.

5.2

5.3

Add weight to Lagrangian, geo

$$\mathcal{L}(\mathbf{W}, b, \alpha, \beta, \xi) = \frac{1}{2}||\mathbf{w}||^2 + C\sum_{i=1}^{m}\xi_i p_i - \sum_{i=1}^{m}\alpha_i[y_i(\mathbf{w}\mathbf{x}_i + b) - 1 + \xi_i] - \sum_{i=1}^{m}\beta_i\xi_i$$

For gradient, (5.26)(5.27)(5.29)(5.30) is same as regular case. (5.28) is replaced with $Cp_i - \alpha_i - \beta_i$. Put them into Lagrangian, get the same result with regular case.

$$\inf \mathcal{L} = \sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\alpha_i\alpha_j y_i y_j \mathbf{x}_i\mathbf{x}_j$$

The constraint condition is $\sum_{i=1}^{m}\alpha_i y_i = 0 \wedge 0 \le \alpha_i, \beta_i \le Cp_i$